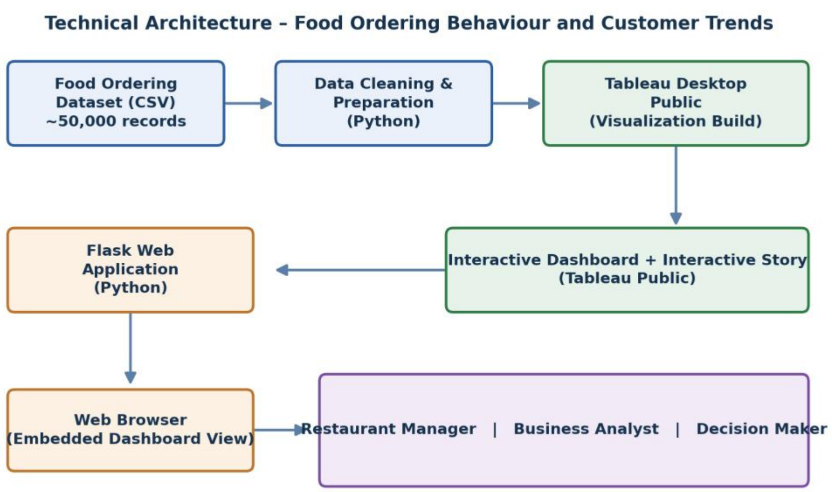
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	8 July 2026
Team ID	
Project Name	Food Ordering Behaviour and Customer Trends
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Food Ordering Behaviour and Customer Trends – Architecture Flow



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How users interact with the analytics application, viewing dashboards through a web browser Web-based Dashboard embedded via the Flask application	HTML5, CSS3, JavaScript
2.	Application Logic-1	Handles HTTP routing, renders web pages, and embeds the Tableau dashboard within the Flask application	Python, Flask
3.	Application Logic-2	Cleans and prepares the raw CSV dataset prior to visualization (handling missing values, formatting fields, etc.)	Python
4.	Application Logic-3	Integrates the Tableau dashboard and story into the web application using an embedded view	Tableau Public (Embedded View), HTML5
5.	Database	This project does not use a traditional database; data is sourced directly from a CSV file	Not Applicable for this project
6.	Cloud Database	Not Applicable for this project	Not Applicable for this project
7.	File Storage	Storage of the raw and cleaned food ordering dataset used to build the dashboard	Local File System (CSV)
8.	External API-1	Not Applicable for this project	Not Applicable for this project
9.	External API-2	Not Applicable for this project	Not Applicable for this project
10.	Machine Learning Model	Not Applicable for this project	Not Applicable for this project
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System Local Server Configuration: Flask development server run on Windows/macOS with Python 3 Version Control: Source code maintained using Git and hosted on GitHub (no cloud server used)	Local Machine, Tableau Public (Dashboard Hosting)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks and libraries used to build the application	Flask (Python), HTML5, CSS3, JavaScript
2.	Security Implementations	Public Tableau dashboard with controlled repository access through GitHub; no sensitive or personal data is exposed in the published dashboard	GitHub repository access control
3.	Scalable Architecture	New dashboards, datasets, and visualizations can be added without changing the overall application architecture	Modular Flask application structure
4.	Availability	Dashboard is accessible through Tableau Public and the Flask web application without dependency on a dedicated hosting server	Tableau Public hosting
5.	Performance	Optimized Tableau visualizations and efficient dashboard loading through pre-aggregated views and filtered data extracts	Tableau Public rendering optimization

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d